A First Course in Abstract Algebra

Part 9: Factorization

Detailed worked solutions

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Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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45 Unique Factorization Domains

Exercise 45.1: An integer prime

A nonzero nonunit is irreducible when every factorization has a unit factor. If $5 = ab$ in $\mathbb{Z}$, then $|a||b| = 5$. Since 5 is prime, one of $|a|, |b|$ is 1. Thus one factor is a unit, and 5 is irreducible.

Exercise 45.2: A negative prime

The units of $\mathbb{Z}$ are $1, -1$. If $-17 = ab$, then $17 = |a||b|$, so one factor has absolute value 1. Thus -17 is irreducible. Equivalently, it is an associate of the irreducible 17.

Exercise 45.3: A composite integer

The factorization $14 = 2 \cdot 7$ uses two nonunits of $\mathbb{Z}$. Therefore 14 is reducible.

Exercise 45.4: A primitive linear polynomial

Suppose $2x - 3 = gh$ in $\mathbb{Z}[x]$. Degrees add, so one factor, say g, is a nonzero integer c. It divides both 2 and -3, hence $c = \pm 1$. It is a unit. The polynomial is nonconstant, so is itself a nonunit; consequently it is irreducible.

Exercise 45.5: A nonprimitive linear polynomial

In $\mathbb{Z}[x]$, $2x - 10 = 2(x - 5)$. The constant 2 is not a unit of $\mathbb{Z}[x]$, and the nonconstant polynomial $x - 5$ is not a unit either. Therefore the polynomial is reducible.

Exercise 45.6: A linear polynomial over the rationals

Every nonzero constant in $\mathbb{Q}[x]$ is a unit. A factorization of a degree-one polynomial has a degree-zero factor, so that factor is a unit. Thus $2x - 3$ is irreducible in $\mathbb{Q}[x]$.

Exercise 45.7: The same expression in a different domain

Although $2x - 10 = 2(x - 5)$, the factor 2 is a unit in $\mathbb{Q}[x]$. Every factorization has a constant factor because the degree is 1. Therefore $2x - 10$ is irreducible in $\mathbb{Q}[x]$. Irreducibility depends on the coefficient domain.

Exercise 45.8: A linear polynomial modulo eleven

The coefficient 2 is nonzero in the field $\mathbb{Z}_{11}$. Thus $2x - 10$ has degree 1 in $\mathbb{Z}_{11}[x]$ and is irreducible by degree addition. Its constant term is congruent to 1, but this does not change its degree.

Exercise 45.9: Associates in three polynomial rings

Associates are unit multiples. In $\mathbb{Z}[x]$ the only units are ± 1, so the only associates are $2x-7$ and $-2x+7$; four different ones do not exist. In $\mathbb{Q}[x]$, four examples are $2x-7, 4x-14, 6x-21, 8x-28$. In $\mathbb{Z}_{11}[x]$ the same four unit multiples are distinct after reducing coefficients: $2x+4, 4x+8, 6x+1, 8x+5$. Distinctness follows also by comparing their leading coefficients.

Exercise 45.10: Factorization over three domains

First extract the integer content:

$$4x^2 - 4x + 8 = 2 \cdot 2\,(x^2 - x + 2).$$

The discriminant of $x^2 - x + 2$ is $1 - 8 = -7$, so it has no rational root. A quadratic over a field is irreducible exactly when it has no root. Gauss's lemma therefore makes this primitive polynomial irreducible in $\mathbb{Z}[x]$ as well. This is the factorization into irreducibles over $\mathbb{Z}$. Over $\mathbb{Q}$, 4 is a unit, and $4(x^2 - x + 2)$ is an irreducible polynomial times a unit. Modulo 11, the roots of $x^2 - x + 2$ are 5 and 7, because $5 + 7 = 1$ and $5 \cdot 7 = 2$ modulo 11. Thus

$$4x^2 - 4x + 8 = 4(x - 5)(x - 7) \quad \text{in } \mathbb{Z}_{11}[x].$$

The two linear factors are irreducible and 4 is a unit.

Exercise 45.11: All gcds of three integers

Euclidean division gives $3250 = 13(234) + 208$, $234 = 208 + 26$, and $208 = 8(26)$. Also $1690 = 65(26)$. Thus the positive gcd is 26. Since the only units of $\mathbb{Z}$ are ± 1, all gcds are 26 and -26.

Exercise 45.12: All gcds with a negative entry

We may replace -1960 by 1960 without changing divisibility. The calculations $1960 = 2(784) + 392$, $784 = 2(392)$, $448 = 392 + 56$, and $392 = 7(56)$ give positive gcd 56. All gcds are $56, -56$.

Exercise 45.13: All gcds of four integers

The divisions $2178 = 5(396) + 198$ and $396 = 2(198)$ give $\gcd(2178, 396) = 198$. Since $792 = 4(198)$ and $594 = 3(198)$, the remaining entries do not change it. The gcds are $198, -198$.

Exercise 45.14: Integer content

The positive gcd of $18, -12, 48$ is 6. Consequently $18x^2 - 12x + 48 = 6(3x^2 - 2x + 8)$. The coefficients $3, -2, 8$ have gcd 1, so the second factor is primitive. Changing the sign of both factors gives the equivalent associate choice.

Exercise 45.15: Content over a field

In $\mathbb{Q}$ every nonzero element is a unit and is a gcd of these coefficients. One valid choice is content 1 and primitive factor $18x^2 - 12x + 48$ itself. For comparison, the choice $6(3x^2 - 2x + 8)$ is also valid. Over a field every nonzero polynomial is primitive: every common nonzero divisor of its coefficients is already a unit.

Exercise 45.16: A polynomial already primitive

The coefficients $2, -3, 6$ have gcd 1, since any common divisor of 2 and 3 is a unit. Thus $2x^2 - 3x + 6 = 1(2x^2 - 3x + 6)$ is the required decomposition in $\mathbb{Z}[x]$.

Exercise 45.17: Content modulo seven

The coefficient ring is the field $\mathbb{Z}_7$. Take content 1 and primitive factor $2x^2 + 4x + 6$. Alternatively choose content 2 and write $2(x^2 + 2x + 3)$, since $2^{-1} = 4$ modulo 7. All nonzero constants are units, so these choices are equivalent.

Exercise 45.18: Definition of associates

The version using a/b is not suitable when $b = 0$. The correct definition for all $a, b \in D$ is: a and b are associates if $a = ub$ for some unit $u \in D$. This includes 0 being an associate of itself. If $b \neq 0$, the quotient criterion in the fraction field is equivalent, provided that quotient is a unit of D.

Exercise 45.19: Definition of irreducible

An irreducible is a nonzero nonunit $p \in D$ such that whenever $p = ab$ with $a, b \in D$, at least one of a, b is a unit. One cannot prohibit every factorization, because $p = 1 \cdot p$ is always available.

Exercise 45.20: Definition of prime

A prime is a nonzero nonunit p such that $p \mid ab$ implies $p \mid a$ or $p \mid b$, for every $a, b \in D$. The expression "smaller" has no intrinsic meaning in an arbitrary integral domain.

Exercise 45.21: Ten statements about factorization

(a) True. In a field there are no nonzero nonunits, so the factorization requirement is vacuous.

(b) True. The only ideals of a field are (0) and (1).

(c) True. The PID factorization theorem proves existence and uniqueness.

(d) False. $\mathbb{Q}[x, y]$ is a UFD, but (x, y) is not principal: a common divisor of x, y is a unit, which would make that ideal the whole ring.

(e) True, by Gauss's theorem applied to the UFD $\mathbb{Z}$.

(f) False. The irreducibles 2 and 3 of $\mathbb{Z}$ are not associates.

(g) False. $\mathbb{Z}$ is a PID but $(2, x)$ is not principal in $\mathbb{Z}[x]$.

(h) True, by the polynomial UFD theorem.

(i) False literally. Associates may replace p: $2 \mid 6$, but $6 = (-2)(-3)$ has no factor literally equal to 2. The correct claim is that an associate of p appears.

(j) True. Being an integral domain is part of the definition of UFD.

Exercise 45.22: Irreducibles in a polynomial UFD

There are two types. Constant irreducibles are precisely the irreducibles of D. A nonconstant $f \in D[x]$ is irreducible precisely when it is primitive and irreducible in $F[x]$. Indeed, a nonunit content would itself give a factorization; after removing content, Gauss's lemma transfers reducibility between $D[x]$ and $F[x]$. Constant nonzero elements become units in $F[x]$, so they must be treated separately.

Exercise 45.23: Irreducible after extending coefficients

Take $D = \mathbb{Z}$, $F = \mathbb{Q}$, and $g(x) = 2x + 2$. It is degree 1, hence irreducible in $\mathbb{Q}[x]$. In $\mathbb{Z}[x]$ the factorization $g = 2(x + 1)$ has two nonunit factors, so it is reducible. Its nonunit content is precisely what disappears on passing to F.

Exercise 45.24: Why zero divisors obstruct factorization

The element $(1, 0)$ is a nonzero nonunit of $\mathbb{Z} \times \mathbb{Z}$, but $(1, 0) = (1, 0)(1, 0)$, a product of two nonunits. More strongly, it cannot be a product of irreducibles. Suppose $(1, 0) = \prod_j (a_j, b_j)$. All a_j are ± 1 and at least one $b_j = 0$. That factor $(\pm 1, 0)$ is reducible because $(\pm 1, 0) = (1, 0)(\pm 1, 0)$. Hence there is no irreducible factorization of $(1, 0)$. The domain hypothesis prevents precisely this kind of idempotent behavior.

Exercise 45.25: A prime is irreducible

Let p be prime and write $p = ab$. Since $p \mid ab$, assume, after exchanging factors if necessary, that $p \mid a$. Write $a = pc$. Then $p = pcb$; cancel the nonzero p in the domain to obtain $1 = cb$. Thus b is a unit. Since a prime is by definition a nonzero nonunit, this proves irreducibility.

Exercise 45.26: An irreducible in a UFD is prime

Suppose $p \mid ab$, so $ab = pc$. If a or b is zero, p divides that factor. Otherwise factor a, b, c into irreducibles, allowing a unit factor when necessary. The concatenated factorizations of a, b and the factorization p followed by that of c represent the same nonzero element. Uniqueness forces an associate of p to occur in the factorization of a or b. It therefore divides that factor, as does p.

Exercise 45.27: Associates form an equivalence relation

Reflexivity follows from $a = 1a$. If $a = ub$ with u a unit, then $b = u^{-1}a$, proving symmetry. If additionally $b = vc$ with v a unit, then $a = (uv)c$, and uv is a unit, proving transitivity. These arguments need no cancellation and work even in a commutative ring with zero divisors.

Exercise 45.28: Products of nonzero nonunits

If a, b are nonzero nonunits, the domain condition gives $ab \neq 0$. If ab were a unit with inverse c, then $a(bc) = 1$ and $b(ac) = 1$, making both factors units, a contradiction. Thus the set is closed. It is not a group: an identity on this nonempty set would satisfy $ea = a$ for nonzero a, hence $e = 1$ by cancellation, but 1 is excluded. If the domain is a field, the set is empty and is still not a group.

Exercise 45.29: A divisor of a primitive polynomial

Write the primitive polynomial as $f = gh$ in $D[x]$. If a nonunit irreducible p divided every coefficient of g, then it would divide every coefficient of $gh = f$. That contradicts the content of f being a unit. Since every nonunit of the UFD has an irreducible divisor, the coefficients of g have no common nonunit divisor; hence g is primitive. The same argument applies to any divisor, constant or nonconstant.

Exercise 45.30: Containment in a maximal ideal

The statement needs the word *proper*: the ideal D itself is not contained in a maximal ideal. For a proper ideal I in a PID, suppose no maximal ideal contains it. Then I is not maximal, so there is a proper ideal I_1 strictly containing I. Repeating produces $I \subsetneq I_1 \subsetneq I_2 \subsetneq \cdots$. The ascending chain condition for a PID forbids this. Thus the process terminates at a maximal ideal containing I.

Exercise 45.31: Factoring a difference of cubes

We have $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. Regard these as polynomials in x over $\mathbb{Q}[y]$. The first is primitive and degree 1, so is irreducible over $\mathbb{Q}(y)$ and therefore over $\mathbb{Q}[y]$. The second is primitive and has discriminant $-3y^2$. This is not a square in $\mathbb{Q}(y)$: otherwise dividing a putative square root by y would give a rational function whose square is -3, impossible in the ordered field $\mathbb{Q}(y)$ (or by comparing coprime numerator and denominator). Thus this quadratic is irreducible over $\mathbb{Q}(y)$; Gauss's lemma gives irreducibility in $\mathbb{Q}[x, y]$.

Exercise 45.32: Three equivalent finiteness conditions

Plan. Prove a cycle of implications for ideals ordered by inclusion.

ACC implies the maximum condition.

If a nonempty collection has no maximal member, choose an ideal in it and then a strictly larger member repeatedly. This contradicts ACC.

The maximum condition implies finite generation.

For an ideal I, consider all ideals generated by finite subsets of I (including the empty generating set). A maximal such ideal J exists. If $a \in I \setminus J$, adjoining a gives a strictly larger finitely generated ideal inside I, a contradiction. Thus $J = I$.

Finite generation implies ACC.

Given $I_1 \subseteq I_2 \subseteq \cdots$, its union I is an ideal: any two elements lie together in a sufficiently late term, and multiplication by ring elements remains in that term. A finite generating set of I lies in some I_N, so $I = I_N$ and all later terms coincide with it. This is ACC.

Exercise 45.33: Descending chains and minimal ideals

As usual the minimum condition must concern *nonempty* collections; an empty collection cannot contain a minimal member. If DCC holds and a nonempty collection had no minimal member, successively choosing a smaller member would produce an infinite strictly descending chain. Conversely, apply the minimum condition to the set of terms of a descending chain. If I_N is minimal among those terms, then for $n \geq N$, $I_n \subseteq I_N$ forces $I_n = I_N$. Thus the chain stabilizes and DCC holds.

Exercise 45.34: ACC without DCC

Use $\mathbb{Z}$. Every ideal is $n\mathbb{Z}$, hence finitely generated, so ACC holds by Exercise 45.32. But

$$2\mathbb{Z} \supsetneq 4\mathbb{Z} \supsetneq 8\mathbb{Z} \supsetneq \cdots$$

is an infinite strictly descending chain: 2^k belongs to the kth ideal but not the next. Therefore DCC fails.

46 Euclidean Domains

Exercise 46.1: The square of absolute value

Yes. This map has nonnegative integer values. Ordinary integer division gives $a = bq + r$ with $0 \le r < |b|$, hence $r = 0$ or $r^2 < b^2$. For nonzero integers a, b, $a^2 \le a^2b^2$ because $b^2 \ge 1$. Both norm axioms hold.

Exercise 46.2: Degree over the integers

No. Divide $a = 1$ by $b = 2$ in $\mathbb{Z}[x]$. A nonzero remainder would have to satisfy $\deg r < \deg 2 = 0$, impossible. A zero remainder would say $1 = 2q$ in $\mathbb{Z}[x]$, also impossible. Degree division works over a field, not over arbitrary integer coefficients.

Exercise 46.3: Leading-coefficient size

No. Take dividend x and divisor 1. Every nonzero polynomial has norm at least 1, so a remainder with norm smaller than that of 1 must be zero; here zero remainder is possible, so this example alone is no obstruction. Instead divide x by x^2; zero remainder is impossible by degrees, while any nonzero remainder has leading-coefficient absolute value at least $1 = v(x^2)$. Thus the required division is impossible.

Exercise 46.4: A rational-valued proposal

No under the stated definition, since a Euclidean norm must take values in the nonnegative *integers*; $v(1/2) = 1/4$ does not. The monotonicity axiom also fails: $v(1) = 1 > v(1 \cdot 1/2) = 1/4$.

Exercise 46.5: A constant norm on a field

Yes. For $b \neq 0$ in $\mathbb{Q}$, set $q = a/b$ and $r = 0$. This gives the division axiom without needing a smaller norm. For nonzero a, b, $v(a) = 50 = v(ab)$ gives monotonicity.

Exercise 46.6: An explicit integer Bezout identity

Back substitution along the Euclidean algorithm gives

$$\begin{aligned} 22471 &= 6(3266) + 2875, & 3266 &= 2875 + 391, \\ 2875 &= 7(391) + 138, & 391 &= 2(138) + 115, \\ 138 &= 115 + 23, & 115 &= 5(23). \end{aligned}$$

Hence $23 = 138 - 115 = 3(138) - 391 = 3(2875) - 22(391) = 25(2875) - 22(3266) = 25(22471) - 172(3266)$. Direct multiplication verifies $561775 - 561752 = 23$, so $\lambda = 25, \mu = -172$.

Exercise 46.7: A larger integer gcd

Successive divisions give

$$\begin{aligned} 49349 &= 3(15555) + 2684, & 15555 &= 5(2684) + 2135,\\ 2684 &= 2135 + 549, & 2135 &= 3(549) + 488,\\ 549 &= 488 + 61, & 488 &= 8(61). \end{aligned}$$

The last nonzero remainder is 61, so a gcd is 61 (and -61 is an associate choice).

Exercise 46.8: Back substitution for the larger gcd

Using the remainders just found,

$$\begin{aligned} 61 &= 549 - 488 = 4(549) - 2135\\ &= 4(2684) - 5(2135) = 29(2684) - 5(15555)\\ &= 29(49349) - 92(15555). \end{aligned}$$

Thus $\lambda = 29, \mu = -92$. Checking the final equality gives $1431121 - 1431060 = 61$.

Exercise 46.9: A polynomial gcd

Let f, g denote the displayed degree-ten and degree-six polynomials. Exact polynomial division gives

$$\begin{aligned} f &= (x^4 - 2x)g + r,\\ r &= -x^4 - 3x^3 - 2x^2 - 5x + 3\\ &= (-x - 3)(x^3 + 2x - 1),\\ g &= (x^3 - 3x^2 + x - 2)(x^3 + 2x - 1). \end{aligned}$$

The cofactors $-x - 3$ and $x^3 - 3x^2 + x - 2$ are relatively prime: substituting $x = -3$ into the latter gives $-59 \neq 0$, so its linear possible common factor does not divide it. Consequently a monic gcd is $x^3 + 2x - 1$. The first division and the displayed factorizations can be checked by multiplying out; they also show directly why no further factor is common.

Exercise 46.10: A gcd of many elements

Compute $d_2 = \gcd(a_1, a_2)$, then $d_3 = \gcd(d_2, a_3)$, and continue until d_n. At each step, an element divides d_k exactly when it divides all $a_1, \ldots, a_k$. This follows by induction from the defining common-divisor property of a gcd. Therefore d_n is a gcd of the entire list. Ignore zero entries, or use $\gcd(a, 0) = a$ up to associates.

Exercise 46.11: Four integers by successive gcds

First $2178 = 5(396) + 198$ and $396 = 2(198)$, so the first gcd is 198. Since $792 = 4(198)$, it remains 198 after the third entry. Finally $726 = 3(198) + 132$, $198 = 132 + 66$, and $132 = 2(66)$. Thus the positive gcd is 66.

Exercise 46.12: The ideal of even constant terms

(a) $\mathbb{Z}[x]$ is a UFD because $\mathbb{Z}$ is a UFD and Gauss's theorem preserves this property on adjoining x.

(b) Let I consist of polynomials with even constant term. Differences preserve this property; multiplying such a polynomial by any integer polynomial multiplies its constant term by an integer, so it remains even. Thus I is an ideal, equal to $(2, x)$.

(c) If $I = (h)$, then h divides both 2 and x. Dividing 2 forces h to be a constant integer divisor of 2; dividing x then forces $h = \pm 1$. This would put 1 in I, contrary to its odd constant term. Hence I is not principal.

(d) A Euclidean domain is a PID. Since this ring is not a PID, it is not Euclidean.

Exercise 46.13: Ten statements about Euclidean domains

(a) True: use a nonzero element of smallest norm in an ideal and divide by it.

(b) False: for example $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is a PID but is not Euclidean. The implication has no converse in general.

(c) True, since every PID is a UFD.

(d) False: $\mathbb{Z}[x]$ is a UFD but not a PID, hence not Euclidean.

(e) True: in the field $\mathbb{Q}$, $1/2$ divides every rational, and every common divisor divides it.

(f) True: successive nonnegative integer norms strictly decrease until the remainder is zero.

(g) True: apply $v(a) \leq v(ab)$ with $a = 1$.

(h) False: in $\mathbb{Z}$, $v(-1) = v(1)$ for $v(n) = |n|$, although $-1 \neq 1$.

(i) True: equality $v(a) = v(1)$ would make a a unit by the norm characterization, proved in Exercise 46.16.

(j) True, with the degree norm and polynomial division over F.

Exercise 46.14: A norm measures, rather than changes, arithmetic

No. Addition, multiplication, units, divisibility, ideals, and gcds are defined by the domain operations. A chosen norm helps find quotients and remainders, but does not alter these operations. Different permissible norms can change which division choices the algorithm makes; the final gcds still differ only by units.

Exercise 46.15: Equal norms for associates

If $a = bu$ with u a unit, monotonicity gives $v(b) \leq v(bu) = v(a)$. Also $b = au^{-1}$, so $v(a) \leq v(b)$. Thus $v(a) = v(b)$. The norms are defined here for nonzero associates.

Exercise 46.16: Strict growth detects a nonunit factor

If b is a unit, a and ab are associates, so their norms are equal. Conversely, suppose $v(a) = v(ab)$. Divide a by ab: write $a = abq + r$, with $r = 0$ or $v(r) < v(ab) = v(a)$. But $r = a(1 - bq)$, so if $r \neq 0$, monotonicity gives $v(a) \leq v(r)$, a contradiction. Thus $r = 0$, and cancellation of $a \neq 0$ yields $1 = bq$. Therefore b is a unit. As $v(a) \leq v(ab)$ always, nonunit b is exactly the strict-inequality case.

Exercise 46.17: The nonunits need not form an ideal

The proposed set is the nonunits together with zero, by the unit characterization. In $\mathbb{Z}$ with absolute-value norm, both 2 and 3 lie in it, but $3 - 2 = 1$ does not. It is therefore not an additive subgroup and is not an ideal in general.

Exercise 46.18: Every field is Euclidean

Define $v(a) = 0$ for every nonzero field element. Divide a by nonzero b by taking $q = ab^{-1}$ and $r = 0$. This satisfies the division axiom, and $v(a) = v(ab) = 0$ satisfies monotonicity. Thus every field is Euclidean.

Exercise 46.19: Changing the scale of a norm

(a) Since $v(a) \geq v(1)$, the assumption $s + v(1) \geq 0$ ensures $\eta(a) \geq 0$. Adding the same integer to two norms preserves both their strict and weak inequalities, so it preserves the two axioms.

(b) Multiplication by a positive integer t preserves nonnegative integer values and all inequalities, so $\lambda(a) = tv(a)$ is a Euclidean norm.

(c) Set $\mu(a) = 1 + 100(v(a) - v(1))$. This is a positive integer norm by (a),(b); it equals 1 on units. A nonunit has integer norm strictly larger than $v(1)$, so $\mu(a) \geq 101 > 100$.

Exercise 46.20: Least common multiples

Let $I = (a) \cap (b)$, the ideal of common multiples. It is nonzero because $ab \in I$ and $ab \neq 0$. A Euclidean domain is a PID, so $I = (c)$ for some nonzero c. Membership of c in both ideals says $a \mid c, b \mid c$. Every common multiple belongs to (c), hence is divisible by c. These are exactly the required lcm properties. Multiplying c by a unit gives another lcm.

Exercise 46.21: Generating the additive integers

The subgroup generated by r, s is $\{ur + vs : u, v \in \mathbb{Z}\}$. Bezout's theorem identifies this with $d\mathbb{Z}$, where $d = \gcd(r, s) > 0$: every combination is divisible by d, and a Bezout combination equals d. It is all $\mathbb{Z}$ precisely when $d = 1$.

Exercise 46.22: A solvable congruence when the coefficient is coprime

If $\gcd(a, n) = 1$, choose integers u, v with $ua + vn = 1$. Multiplying by b gives $a(ub) + n(vb) = b$. Thus $x = ub$ satisfies $ax \equiv b \pmod{n}$. In $\mathbb{Z}_n$ this says the nonzero residue of a is invertible when it is coprime to n.

Exercise 46.23: The general congruence criterion

Put $d = \gcd(a, n) > 0$. If $ax \equiv b \pmod{n}$, then $b = ax - kn$ for some integer k, and $d \mid b$. Conversely, if $b = dc$, multiply a Bezout identity $ua + vn = d$ by c. It gives $a(uc) \equiv b \pmod{n}$. In the residue ring $\mathbb{Z}_n$, the principal ideal generated by $\bar{a}$ consists exactly of residues represented by multiples of d; solvability asks whether $\bar{b}$ lies in this ideal.

Exercise 46.24: Constructing a solution

Run the extended Euclidean algorithm to obtain $d = ua + vn$. If $d \nmid b$, no solution exists. Otherwise $x_0 = u(b/d)$ is a solution; all solutions are $x = x_0 + k(n/d)$. For $22x \equiv 18 \pmod{42}$, $d = 2$ and $2 = 2(22) - 42$. Thus $x_0 = 2(18/2) = 18$. Check: $22(18) - 18 = 378 = 9(42)$. All integer solutions are $x = 18 + 21k$; modulo 42 these are 18 and 39.

47 Gaussian Integers and Multiplicative Norms

Exercise 47.1: Splitting five

In $\mathbb{Z}[i]$, $5 = (2+i)(2-i)$. Both factors have norm 5, a rational prime, so a factorization of either would force one factor to have norm 1 and hence be a unit. Both are irreducible.

Exercise 47.2: Seven stays irreducible

The norm of 7 is 49. If $7 = \alpha\beta$ with both factors nonunits, their norms must both be 7. But $a^2 + b^2 = 7$ has no integer solutions: the possible squares at most 7 are 0,1,4 and none sum to 7. Thus such a factorization is impossible, and 7 is irreducible in $\mathbb{Z}[i]$.

Exercise 47.3: Factoring a norm twenty-five element

A direct multiplication gives $4 + 3i = i(2-i)^2$. Indeed $(2-i)^2 = 3 - 4i$ and multiplying by i gives $4 + 3i$. The factor i is a unit; $N(2-i) = 5$ is prime, so $2 - i$ is irreducible.

Exercise 47.4: Factoring a norm eighty-five element

We have $6 - 7i = (2+i)(1-4i)$. Expanding gives $2 - 8i + i - 4i^2 = 6 - 7i$. The norms are 5 and 17, respectively, both rational primes, so both factors are irreducible.

Exercise 47.5: Failure of unique factorization

Put $s = \sqrt{-5}$. Then $6 = 2 \cdot 3 = (1+s)(1-s)$. The norm $N(a+bs) = a^2 + 5b^2$ is multiplicative, and its units have norm 1. There are no elements of norm 2 or 3. Thus 2 (norm 4), 3 (norm 9), and $1 \pm s$ (norm 6) cannot split into two nonunits: their proper possible norm factors would require norm 2 or 3. All four displayed factors are irreducible. Their norms $4, 9, 6, 6$ show neither 2 nor 3 is associate to either $1+s$ or $1-s$. Hence the two factorizations are different up to associates.

Exercise 47.6: One Gaussian division

Compute

$$\frac{7+2i}{3-4i} = \frac{(7+2i)(3+4i)}{25} = \frac{13+34i}{25}.$$

Round both coordinates to nearest integers to choose $\sigma = 1 + i$. Then $(3-4i)(1+i) = 7 - i$, so $\rho = 3i$. We have $7 + 2i = (3-4i)(1+i) + 3i$, and $N(3i) = 9 < 25 = N(3-4i)$.

Exercise 47.7: A Gaussian gcd

The divisions

$$8 + 6i = i(5 - 15i) + (-7 + i), \qquad 5 - 15i = (-1 + 2i)(-7 + i)$$

have first remainder norm 50, smaller than 250. The next remainder is zero. Thus the last nonzero remainder $-7 + i$ is a gcd. Its associates, for example $7 - i$, are equally valid.

Exercise 47.8: Ten statements about norms

(a) True. The Gaussian integer norm makes the ring Euclidean, hence a PID.

(b) True. Round real and imaginary coordinates in Gaussian division.

(c) True: $n = n + 0i$.

(d) False: $1/2$ has no integer real coordinate.

(e) True, since the nonnegative remainder norms decrease.

(f) True. Norm divisibility restricts possible factor norms, as in Exercises 47.1–47.4.

(g) True. From $N(u)N(u^{-1}) = N(1) = 1$, integer norms give $|N(u)| = 1$.

(h) False. Degrees add rather than multiply; also the norm of 1 would be 0 rather than 1.

(i) True. For nonzero polynomials, degree addition gives $2^{\deg(fg)} = 2^{\deg f}2^{\deg g}$; declaring $N(0) = 0$ handles zero products and $N(1) = 1$.

(j) True. It is a subring of $\mathbb{C}$, hence a domain; Exercise 47.5 gives nonunique factorization.

Exercise 47.9: Smallest nonunit norm

Suppose $\pi = ab$ with both factors nonunits. Since $\pi \neq 0$ and the ring is a domain, $a, b \neq 0$. By the hypothesis relating norm and units, both $|N(a)|, |N(b)|$ are integers greater than 1. Minimality gives $|N(a)| \geq |N(\pi)|$. Multiplicativity then gives $|N(\pi)| = |N(a)||N(b)| > |N(a)| \geq |N(\pi)|$, a contradiction. Thus every factorization has a unit factor, proving irreducibility.

Exercise 47.10: Which integer primes split

(a) $2 = -i(1 + i)^2$, and $1 + i$ is irreducible because its norm is the prime 2.

(b) If $p \equiv 1 \pmod 4$, the sum-of-two-squares theorem gives $p = a^2 + b^2 = (a + bi)(a - bi)$ with both factors nonunits. If an odd prime $p \equiv 3 \pmod 4$ were reducible, both nonunit factor norms would equal p. But an integer square is 0 or 1 modulo 4, so $a^2 + b^2$ cannot be 3 modulo 4. This excludes norm p and proves irreducibility. Every odd prime is 1 or 3 modulo 4, completing both directions.

Exercise 47.11: The Gaussian norm identity

For $\alpha = a + bi$, $\beta = c + di$, multiplication gives $\alpha\beta = (ac - bd) + (ad + bc)i$. Therefore

$$\begin{aligned} N(\alpha\beta) &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 \\ &= (a^2 + b^2)(c^2 + d^2) = N(\alpha)N(\beta). \end{aligned}$$

The cross terms cancel. Since $N(\alpha) = a^2 + b^2$ is a sum of squares, it is nonnegative. Also $N(\alpha) = 0$ exactly for $a = b = 0$, and $N(1) = 1$, giving the stated norm properties.

Exercise 47.12: A multiplicative norm with square root minus five

Write $\alpha = a + b\sqrt{-5}$ and $\beta = c + d\sqrt{-5}$. Their product is $(ac - 5bd) + (ad + bc)\sqrt{-5}$. Expanding gives

$$(ac - 5bd)^2 + 5(ad + bc)^2 = (a^2 + 5b^2)(c^2 + 5d^2).$$

Thus $N(\alpha\beta) = N(\alpha)N(\beta)$. The values are nonnegative integers, $N(0) = 0$, and $N(1) = 1$.

Exercise 47.13: Existence of irreducible factorizations from a norm

Use strong induction on the positive integer $m = |N(a)| > 1$. If a is irreducible we are done. Otherwise $a = bc$ with b, c nonunits, necessarily nonzero. Their absolute norms are integers greater than 1 and multiply to m, so both are strictly smaller than m. By induction both factors are products of irreducibles; concatenate these factorizations. The smallest possible nonunit norm supplies the base case, since it cannot split as a product of two smaller norms greater than 1.

Exercise 47.14: Another Gaussian gcd

Gaussian division yields

$$16 + 7i = (1 + i)(10 - 5i) + (1 + 2i), \qquad 10 - 5i = (-5i)(1 + 2i).$$

The first remainder has norm 5, below $N(10 - 5i) = 125$, and the second remainder is zero. Thus $1 + 2i$ is a gcd, up to multiplication by $1, -1, i, -i$.

Exercise 47.15: Finite Gaussian residue rings

(a) Division by nonzero α gives each residue class a representative r with $N(r) < N(\alpha)$. There are finitely many integer lattice points in this bounded disk, so there are finitely many residue classes. More exactly the quotient has $N(\alpha)$ elements: multiplication by $a+bi$ has integer lattice matrix $\left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$ of determinant $a^2 + b^2$.

(b) An irreducible in this PID is prime, so its principal ideal is maximal. Its quotient is therefore a field.

(c) For (3), there are $N(3) = 9$ residues and characteristic 3. For $(1 + i)$, there are 2 residues and characteristic 2 because $2 = -i(1 + i)^2$. For $(1 + 2i)$, there are 5 residues and characteristic 5, since $5 = (1+2i)(1-2i)$. These are fields because the indicated divisors are Gaussian irreducibles (7-like reasoning for 3; prime norm for the other two).

Exercise 47.16: Imaginary quadratic norms

Let $s = \sqrt{-n}$, with n positive and square free.

(a) Multiplication and expansion give $N((a + bs)(c + ds)) = (ac - nbd)^2 + n(ad + bc)^2 = (a^2 + nb^2)(c^2 + nd^2)$. Values are integers, with $N(1) = 1$ and $N(\alpha) = 0$ only for $\alpha = 0$.

(b) If α is a unit, multiplicativity forces $N(\alpha) = 1$. Conversely, if $N(\alpha) = 1$, then $\alpha\bar{\alpha} = 1$, and $\bar{\alpha} = a - bs$ is in the same ring, giving an inverse.

(c) Every nonzero nonunit has integer norm greater than 1. Apply the strong-induction proof of Exercise 47.13 to obtain a factorization into irreducibles. This proves existence, not uniqueness.

Exercise 47.17: Real quadratic norms

For the intended real quadratic case assume the square-free integer $n > 1$. The representation $a+b\sqrt{n}$ is then unique because $\sqrt{n}$ is irrational. Define $N(a + b\sqrt{n}) = a^2 - nb^2$, allowing negative values.

(a) Direct expansion gives $(ac+nbd)^2-n(ad+bc)^2 = (a^2-nb^2)(c^2-nd^2)$, so the norm is multiplicative. A nonzero element has nonzero norm by irrationality of $\sqrt{n}$.

(b) The appropriate statement is $|N(\alpha)| = 1$ if and only if α is a unit. A unit has integer norm dividing 1. Conversely, for norm ± 1, $\alpha^{-1} = \bar{\alpha}/N(\alpha)$ lies in the ring. Requiring $N = 1$ literally would be false, e.g. $1 + \sqrt{2}$ has norm -1 and inverse $\sqrt{2} - 1$.

(c) The absolute norm is a positive integer, greater than 1 on nonzero nonunits. The induction in Exercise 47.13 therefore proves existence of an irreducible factorization.

The original reference includes square-free $n = 1$; here the expression $a^2 - b^2$ is not well defined on $a + b\sqrt{1}$ (1 has representations $(1, 0)$ and $(0, 1)$). Thus that degenerate case must be excluded for this norm construction.

Exercise 47.18: Euclidean division with square root minus two

For nonzero β, write $\alpha/\beta = u + v\sqrt{-2}$ with $u, v \in \mathbb{Q}$. Choose integers m, k with $|u - m| \leq 1/2$ and $|v - k| \leq 1/2$. Put $q = m + k\sqrt{-2}$ and $r = \alpha - \beta q$. Then

$$N(r) = N(\beta)\big((u - m)^2 + 2(v - k)^2\big) \leq \tfrac{3}{4}N(\beta) < N(\beta).$$

If $r = 0$ the required alternative holds as well. All nonzero norms are positive integers and multiplication multiplies them by an integer at least 1, so the second Euclidean axiom holds too. Hence $\mathbb{Z}[\sqrt{-2}]$ is Euclidean.